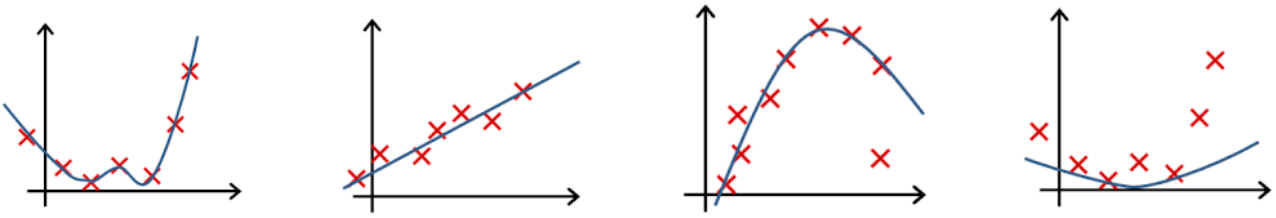
Tutorial Questions | Week 2

COSC2779 - Deep Learning

This tutorial is aimed at reviewing basic machine learning concepts. Please try the questions before you join the session.

1. A computer program is said to learn from experience E with respect to some task T and some performance measure P if its performance on T, as measured by P, improves with experience E. Suppose we feed a learning algorithm a lot of historical weather data, and have it learn to predict weather.
   (a) What would be a reasonable choice for P?
   (b) What is T?
2. Suppose you are working on stock market prediction, and you would like to predict the price of a particular stock tomorrow (measured in dollars). You want to use a learning algorithm for this. Would you treat this as a classification or a regression problem?
3. In which one of the following figures do you think the hypothesis has overfit the training set?



(a) (b) (c) (d)

4. What's the trade-off between bias and variance?
5. How do you ensure you're not overfitting with a model?
6. We can regularize a regression model by introducing a ridge penalty as shown in the equation below:

   $\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}(y - \hat{y})^2 + \lambda w^\top w$

   what happens if we tune λ by looking at the performance on the train set?
7. The breast cancer dataset is a standard machine learning dataset. It contains 9 attributes describing 286 women that have suffered and survived breast cancer and whether or not breast cancer recurred within 5 years. It is a binary classification problem. Of the 286 women, 201 did not suffer a recurrence of breast cancer, leaving the remaining 85 that did. How would you handle an imbalanced dataset?
8. What validation technique would you use on a time series dataset?